

# Rakesh Pandey

## PRINCIPAL AI & DATA ARCHITECT

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EDUCATION

- Master of Data Science

June 2022

The University of British Columbia (UBC) • Grade: A+, 4/4 • Vancouver

  - Worked with **TRIUMF** and the CERN lab on NA62 experiment to develop a **deep learning classifier** capable of identifying pion, muon, and positron particles with at least 95% pion efficiency while keeping muon to pion misclassification rate less than 0.1. PyTorch was used to implement a customized Point Net and Dynamic Graph CNN.
- Bachelor of Technology, Computer Science and Engineering

July 2014

Karunya University, India • 8.3 out of 10 GPA

  - Received “Best Project Award” for my project “Web Based Penetration Testing Toolkit”, <https://github.com/veerupandey/Penetration-Testing-Toolkit>

SKILLS

AI & ML ENGINEERING

Large Language Models (LLMs)

Agentic Workflows

MLOps/LLMOps

Computer Vision

Deep Learning

PyTorch

MLFlow

PySpark

Databricks

Predictive Modeling

DATA ENGINEERING

GCP

Azure

AWS

Data Lakes

Kafka

SQL

Python

Apache Spark

Airflow

CI/CD

API Design

Docker

PROFESSIONAL EXPERIENCE

- Cofounder & Principal AI Engineer

January 2025 – Present

FirstRatio Intelligence, Canada • [Detailed Projects Portfolio](#)

  - Engineering Leadership:** Led a team of software and AI/ML engineers to design, build, and deploy custom agentic solutions, advanced MLOps platforms, and enterprise data workflows for holding companies and startups.
  - Music Distribution Platform:** Architected a complex conversational agent system that automates music optimization, validates metadata, runs automated copyright verification, and coordinates cross-platform streaming distribution.
  - PE/M&A Due Diligence System:** Engineered an agentic PE Data Room RAG system with a Planner-Generator-Evaluator harness that differentially audits deal memos against Excel models and source PDFs.
  - Agent Fleet & Memory Infrastructure:** Deployed an 8-node distributed Tailscale agent fleet (Slot-Map) managed by a secure custom control plane and powered by self-hosted relational memory graphs.
- Team Lead - AI and ML Engineering

January 2023 – Present

ICBC (Insurance Corporation of British Columbia), Canada

  - Led a team of AI/ML engineers to design and deploy an enterprise-grade AI fraud detection model to replace a legacy commercial solution, significantly increasing detection accuracy and reducing false-positive claims.
  - Spearheaded a team to design and develop an interactive fraud visualizer enabling claims investigators to trace, analyze, and map complex fraud patterns and network links.
  - Designed an AI agent liability resolution predictor utilizing agentic workflows to assist claim adjusters in automatically determining liability during the First Notice of Loss (FNOL) phase.
  - Developed a multi-agent system to aggregate and summarize unstructured claim notes, customer conversation checkpoints, medical records, and other auxiliary documentation, streamlining claims adjudication.
  - Built a dashboard-to-agent pipeline where selecting anomalous data points on a Tableau KPI dashboard triggers automated agents with cross-dataset access to perform deep-dive root-cause investigations.
  - Implemented a computer vision vehicle damage assessment pipeline (Mitchell-style) that automatically analyzes damage photos to estimate repair costs and assess damage severity.
- ML Architect

November 2022 – January 2023

Wavicle Data Solutions

  - Led Databricks adoption for a client to optimize and streamline machine learning workflows.

## Manager Data Engineering

July 2022 – November 2022

Publicis Sapient, Canada

- Retail Media Network - Worked with two retail clients in US to understand current state of AdTech and MarTech and suggested future data pipelines and solution architecture on Google Cloud Platform in order to create an integrated and seamless platform to manage omni-channel campaigns with better measurements and engagement.

## Senior Analyst

January 2019 – August 2021

Boston Consulting Group, Gurgaon, India

- For a leading retailer, led a team to modernize the existing legacy data solution and created a suite of digital commercial tools to help buyers and category managers on Azure using Tableau, PySpark, Azure Databricks, and Azure Data Factory.
- For an airline client, worked on building a cloud native performance revenue tracking system to track revenue at various level of granularity using Azure Data Factory, Spark, Databricks, Azure Synapse. Received "Client Impact Award" for bringing next phase of project to the team.
- Led a team of Data Engineers and Scientists to build a customer 360 (C360) project for one of the largest telecom operators in SEA to provide a 360-degree customer view to the business and data science teams using AWS S3, Glue, Athena etc. Featured as "Most Trusted engineer" in BCG GAMMA newsletter.
- Developed a set of ML models for a telecom client to recommend offers based on customer lifetime management in order to increase prepaid subscriber base revenues. This results in a net revenue increase of 4% for the target group. Received "Project Excellence Award". Solution was build using Python, Spark and MLFlow on Azure Databricks.

## Data Developer

March 2017 – December 2019

Dunnhumby India, Gurugram, India

- Helped in the modernization of an on-premises SAS data pipeline to the Google cloud using Dataproc (Pyspark, Hive), Big Query, and Cloud Composer (Airflow). Created a generic framework for ingesting, transforming, and analyzing retail data.
- Contributed to designing/building CI/CD pipeline for running unit, integration, and behave tests.
- Worked with Data Scientists and Business to move existing analytical solutions and machine learning models to Google Cloud.
- Received "Data Expert" for my outstanding contribution to the project.

## Data Scientist

March 2016 – March 2017

Everlytics Data Science LLP, Bengaluru, India

- Designed, developed, and maintained a pipeline for consuming terabytes of log data per day in near real time from hundreds of machines for an Internet Archive using Lua, Kafka, Elasticsearch, Python, Influx DB, Grafana.
- Contributed to the development of "**Predictive Maintenance Framework**", which is used to detect machine failure and recommend corrective actions for a variety of clients. Developed machine learning models to detect failures before they occur.

## Software Engineer (Business Intelligence)

October 2014 – February 2016

Capgemini India, Mumbai, India

- Built a new module for a German telecom client which includes data flow from multiple customer support system and product sales team, to provide a complete overview of a customer request and timely resolution of complaints using Shell Scripting, Oracle SQL/PL-SQL and IBM Cognos.
- Received "Rising Star Award" for the extra ordinary performance on the project.